

Adaptive Resource Placement for Cyber Defense

Better Protection Through
Multi-Agent Reinforcement Learning

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Project Defense: September 2025

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- **Problem:** Static cyber defenses fail against adaptive adversaries
- **Approach:** Multi-agent reinforcement learning for adaptive deception
- **Solution:** PPO-based agent learning optimal honeypot placement
- **Result:** **89% performance improvement** over baselines



Ancient War Strategy Quotes

Sun Tzu Quotes

- “All warfare is based on deception.”
- “He will win who knows when to fight and when not to fight.”
- “The greatest victory is that which requires no battle.” — Sun Tzu, The Art of War.

These quotes support our approach: creating uncertainty and choosing when to act are central to the defense agent's strategy.

Key Terms & Concepts

Reinforcement Learning Learning by trial and reward feedback.

Agent The automated decision-maker (here: Blue agent).

Environment The simulated network in which agents act.

Policy Mapping from what the agent sees to actions.

Reward Numeric signal

PPO A stable policy-optimisation algorithm used here.

Action What the agent does (deploy/remove/wait).

Observation / State What the agent perceives about the network.

Honeypot / Decoy Fake asset used to lure and study attackers.

Red / Blue Team Red = attacker

The Cybersecurity Challenge

Critical Challenges:

- **Alert Fatigue:** 4,000+ daily alerts, 67% ignored
- **Asymmetric Warfare:** Attackers need one success
- **Static Defenses:** Predictable patterns exploited
- **Base-Rate Fallacy:** Real threats missed

Attacker

Need 1 success

VS

Defender

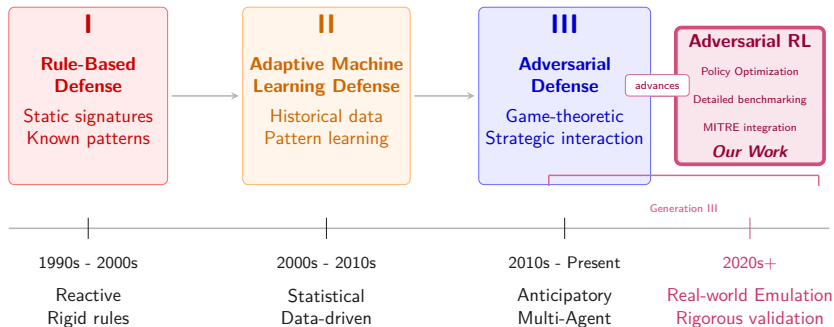
Protect everything

Asymmetric Challenge

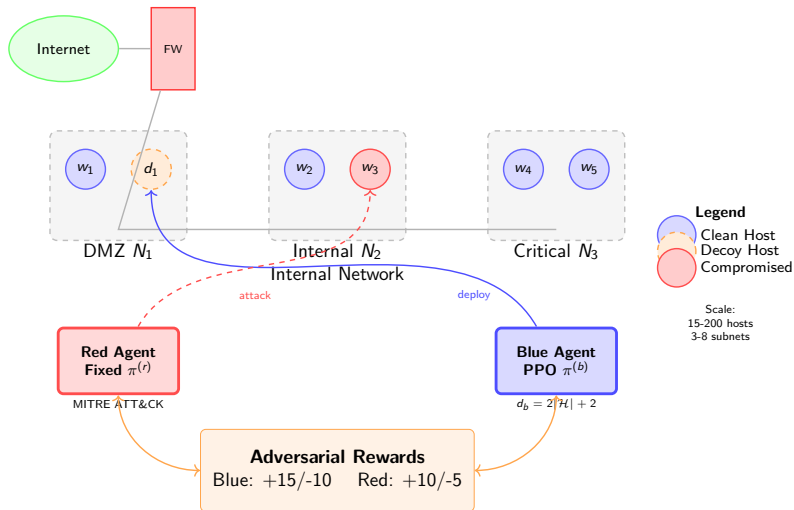
Our Solution

Adaptive AI-driven deception that creates uncertainty for attackers through learned strategic honeypot placement

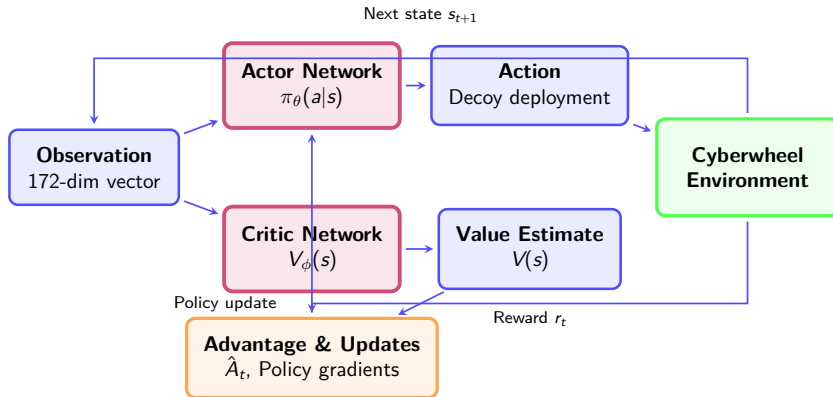
Evolution of Cyber Defense



Network and Agent Environment Overview



PPO Training Architecture



Key Features

- **172-dimensional observation space:** Alert history + network state + metadata
- **Actor-critic framework:** Stable policy optimization with value function guidance

RL Foundations (PPO — concise)

$\pi_{\theta}(a|s)$ (actor / policy)

$V_{\phi}(s)$ (critic / value)

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right]$$

$$L(\theta, \phi) = -L^{\text{CLIP}}(\theta) + c_1 (V_{\phi}(s_t) - R_t)^2 - c_2 \mathcal{H}(\pi_{\theta})$$

$$\text{GAE (compact): } \hat{A}_t = \sum_{l=0}^{k-1} (\gamma \lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V_{\phi}(s_{t+1}) - V_{\phi}(s_t)$$

Notes: clipping stabilises policy updates; value loss (c_1) and entropy regulariser (c_2) improve convergence.

Key Results: 89% Performance Improvement

Algorithm Performance:

Algorithm	Reward
PPO (Ours)	309.14
Static Baseline	163.81
Rule Baseline	79.16
Inactive	-1.61

Security Effectiveness:

- **89% improvement** over static baseline
- **73.2% deception success rate** (vs 45.1% static)
- **82.4% asset protection rate** (vs 61.7% static)
- **2.3 timestep** detection latency (vs 4.1 static)
- **92.3% correlation** with MITRE ATT&CK patterns
- **84.8% defense success** against real attacks

Statistical Validation

ANOVA $p < 0.001$, Cohen's $d > 0.8$, 95% confidence intervals, 5-seed validation

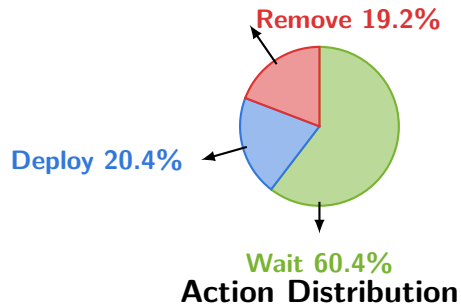
Learned Strategic Behaviors

Emergent Intelligence:

- **Balanced Strategy:** 20.4% deploy, 19.2% remove, 60.4% wait
- **Learned Restraint:** Avoids over-deployment traps
- **Superior Consistency:** $\sigma = 37.37$ vs baseline variance
- **Adaptive Timing:** Strategic waiting periods

Strategic Insights:

- Agent learns **when NOT to act** (patience)
- Balances resource costs vs security benefits
- Adapts to attacker patterns without explicit programming



Scalability Analysis

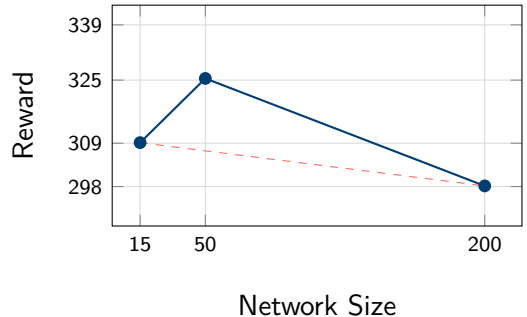
Network Size Performance:

Hosts	Reward	Time
15	309.14	2.5h
50	325.42	6.2h
200	298.17	18.4h

Scalability Properties:

- **Consistent Performance:** 298-325 reward range
- **Sub-quadratic scaling:** $O(n \log n)$ complexity
- **Linear memory usage:** Practical for enterprise
- **Strategic consistency:** Behaviors stable across scales

Performance vs Network Size



Enterprise Ready: Validated up to 200 hosts with maintained effectiveness

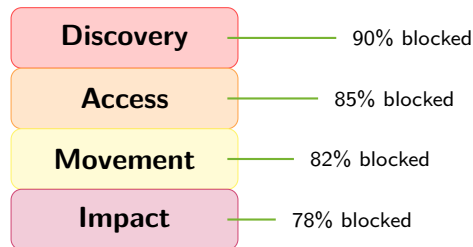
Real-World Validation

MITRE ATT&CK Integration:

- **295 attack techniques** implemented
- **92.3% behavioral correlation** with real patterns
- **Complete kill chain** coverage (4 phases)
- **Atomic Red Team** validation

Attack Defense Success:

- **84.8% success rate** against real attack commands
- **Early disruption:** 2.3 timestep detection
- **Proactive defense:** Uncertainty creation vs reactive response



Attack Phase Defense

Practical Deployment Ready

Experimental Framework:

- **Multi-Agent Setup:** Learning blue vs fixed red (295 techniques)
- **Networks:** 15-200 hosts, realistic enterprise topologies
- **Training:** 10M timesteps, 5 random seeds
- **Baselines:** Static, Rule-based, Random, Inactive

Statistical Rigor:

- **ANOVA:** $p < 0.001$ significance
- **Effect Size:** Cohen's $d > 0.8$ (large)
- **Confidence:** 95% intervals

Novel Contributions:

- 1 **First systematic PPO validation** for cyber deception
- 2 **Real-world MITRE integration** with 92.3% fidelity
- 3 **Scalability characterization** across enterprise networks
- 4 **Strategic behavior analysis** revealing emergent intelligence
- 5 **Reproducible methodology** with open protocols

Research Questions Answered:

- PPO effectively learns adaptive strategies
- 89% improvement over traditional

Future Directions & Impact

Research Extensions:

- Real-world deployment in operational environments
- Adversarial co-evolution with learning attackers (self-play)
- Multi-defender coordination across distributed networks
- Integration with existing security infrastructure (SIEM, EDR)
- Explainable AI for analyst understanding and trust

Technical Advances:

- Advanced reward shaping for complex scenarios
- Transfer learning across network topologies
- Real-time adaptation to novel attack patterns

Broader Impact

This work establishes reinforcement learning as a viable approach for automated cybersecurity defense with rigorous experimental validation.

Key Takeaways

- AI successfully learns sophisticated adaptive defense strategies
- **89% improvement** demonstrates clear superiority
- Strategic restraint emerges naturally without explicit programming
- Performance scales robustly to enterprise networks
- Real attack correlation validates deployment readiness

Adaptive AI-Driven Cyber Defense Works

Performance

89%

Improvement
over static baselines

Intelligence

**Strategic
Behaviors**
emerge naturally

Scale

**Enterprise
Ready**
15-200 hosts validated

Research Impact

First rigorous validation of reinforcement learning for cybersecurity deception with **real-world attack correlation** and **enterprise-scale deployment readiness**

Questions?

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